Cross-Variety Initialization and Alignment-Supervised Durations for One-Hour Hakka Speech Synthesis

Chien-Chih Fang
Master Program in Intelligent Computing and Big Data
Chung Yuan Christian University
Taoyuan, Taiwan
tom30112@gmail.com

I Elizabeth Cha*
Master Program in Intelligent Computing and Big Data
Chung Yuan Christian University
Taoyuan, Taiwan
icha@cycu.edu.tw
*Corresponding author

Shin-Shin Kao
Department of Applied Mathematics
Chung Yuan Christian University
Taoyuan, Taiwan
shinshin@cycu.edu.tw

***Abstract*—Hakka is a tonal language with several regional varieties, many of which have limited recorded speech resources. This paper investigates neural text-to-speech (TTS) for two Taiwanese Hakka varieties, Sixian and Hailu, under a one-hour target-variety training condition. With limited training data, learned alignments may result in inaccurate segment durations and unreliable tone realization. This paper presents XAD-TTS, a Matcha-TTS training recipe that combines cross-variety initialization with forced-alignment duration supervision and an enlarged duration predictor. The target model is initialized from a model trained on a related Hakka variety. Across three training seeds, XAD-TTS increases the proportion of checked-syllable vowel nuclei whose durations, measured on synthesized waveforms, are within 30 ms of the corresponding natural durations from 71.0% to 81.3% for Sixian and from 69.4% to 82.2% for Hailu. For this duration measure, the results are close to those of full-data baselines trained on 48.0 and 47.2 hours of target-variety speech, respectively. XAD-TTS also reduces mel-cepstral distortion by 0.33 dB for Sixian and 0.24 dB for Hailu, while improving objective tone-classification accuracy. An additional automatic speech recognition analysis shows improved agreement for checked-syllable codas and tone marks relative to the transcripts of natural recordings.**

***Index Terms*—duration modeling, forced alignment, Hakka, low-resource languages, speech synthesis, transfer learning**

I. Introduction

Hakka is a Sinitic tonal language spoken in several regional varieties. In Taiwan, Sixian and Hailu are two major Hakka varieties. They share many consonants and vowels but differ in their tone systems. Although Hakka speech technology has begun to emerge [1], recorded speech resources remain limited for individual varieties, making it difficult to develop TTS systems when only a small amount of target-variety data is available.

Neural TTS models such as Matcha-TTS [2] learn alignments between input symbols and acoustic frames during training, typically using monotonic alignment search (MAS) [3]. A duration predictor is then trained using the resulting alignments. Under limited-data conditions, alignment and duration prediction may be less reliable. In Hakka, this issue is particularly relevant to checked syllables, which end in an unreleased stop, written as *-b*, *-d*, or *-g* in the romanization used in this study. These syllables typically have relatively short vowel nuclei; therefore, duration is one of the acoustic cues relevant to their realization.

Previous studies have explored approaches to low-resource TTS, including initialization from models trained on related, better-resourced languages [4], [5], and the use of durations obtained from an external aligner, as in FastSpeech 2 [6] with the Montreal Forced Aligner (MFA) [7]. This paper combines cross-variety initialization and alignment-supervised duration modeling for Hakka TTS, focusing on conditions with limited target-variety training data. The main contributions are as follows:

- **XAD-TTS:** XAD-TTS, short for cross-variety initialization with alignment-supervised durations, is a Matcha-TTS training recipe that initializes a target-variety model from a related source variety and supervises an enlarged duration predictor using forced-alignment durations. The aligner is not required during synthesis, although it is used to obtain training targets and to measure durations during evaluation.
- **Experiments across data sizes:** Evaluation on Sixian and Hailu using three training seeds at 1 h and 10 h, together with single-seed full-data baseline references at 48.0 h and 47.2 h.
- **Evaluation beyond spectral distance:** Analysis of synthesized vowel durations, objective tone classification, and ASR-based agreement between synthesized and natural speech.

II. Method

A. Baseline model

The baseline system, denoted Base, uses the original Matcha-TTS architecture [2] and is trained from scratch.

Matcha-TTS encodes input symbols into per-symbol features and predicts their durations. The features are expanded according to the durations, and a conditional flow-matching decoder [8] generates a mel spectrogram. A HiFi-GAN vocoder [9] then converts the mel spectrogram into a waveform.

During training, MAS estimates the alignment between input symbols and mel frames, and the duration predictor is trained using the resulting durations. The training objective consists of duration, prior, and flow-matching losses. In the original inference procedure, predicted durations are rounded up to whole frames. The rounding procedure considered in this study is described in Section II-D.

B. Cross-variety initialization

A source model is trained from scratch on the full training pool of the other Hakka variety: 48.0 h for Sixian or 47.2 h for Hailu. Sixian and Hailu are closely related varieties, and both use a shared symbol table. For target-model initialization, parameters with matching names and shapes are copied from the source model, except for the speaker embeddings and, in XAD-TTS, the enlarged duration predictor. These components are re-initialized, no parameters are frozen, and the optimizer is restarted.

Because the source model contains mel-normalization statistics from the source variety, synthesis uses the statistics calculated from the target-variety training data.

C. Alignment-supervised durations

For each variety, an MFA acoustic model was trained on all HAT TTS recordings of that variety and their transcripts, using a pronunciation lexicon derived from the romanization by deterministic rules. The aligner uses only audio and transcripts, never the TTS model, and the same alignments provide the natural reference durations used in Section III-C. Over 99% of the utterances were successfully aligned; the remaining 42 Sixian and 143 Hailu utterances were excluded from all subsets. The resulting phone intervals are converted into frame counts using a hop size of 256 samples at a sampling rate of 22.05 kHz, corresponding to approximately 11.6 ms per frame.

Matcha-TTS inserts blank tokens between input symbols. Aligned pauses are assigned to the corresponding blank tokens, and each blank receives at least one frame, taken from the preceding phone. The resulting durations replace MAS-derived durations during training. They serve as targets for the duration loss and determine the feature expansion used in the prior and decoder losses.

D. Larger duration predictor and rounding

To increase the capacity of the duration predictor, its kernel size is changed from 3 to 5 and its number of channels from 256 to 512. When cross-variety initialization is applied in XAD-TTS, the enlarged duration predictor and speaker embeddings are re-initialized.

Rounding every predicted duration up can increase the total predicted duration. To reduce the potential effect of rounding, each trained model uses either rounding up or cumulative rounding (rounding the running sum, which preserves the total length). The rule is chosen on the validation set, before any test sentence is synthesized: the rule whose rounded vowel-nucleus durations have the smaller mean absolute error with respect to the MFA-derived durations, averaged over checked and other syllables, is selected. The same selection procedure is applied to all systems, including Base.

XAD-TTS combines the approaches described in Sections II-B to II-D. Its name refers to cross-variety initialization (X) and alignment-supervised durations (AD), the latter including the enlarged predictor and the rounding selection. Fig. 1 illustrates the training, synthesis, and evaluation procedures. The target model is initialized from the source-variety model and trained on the one-hour target training speech and text, while MFA-derived durations from the same data provide supervision for the enlarged duration predictor and alignment information for training. After training, the model synthesizes speech from test text. The synthesized speech is then evaluated using natural test speech as a reference.

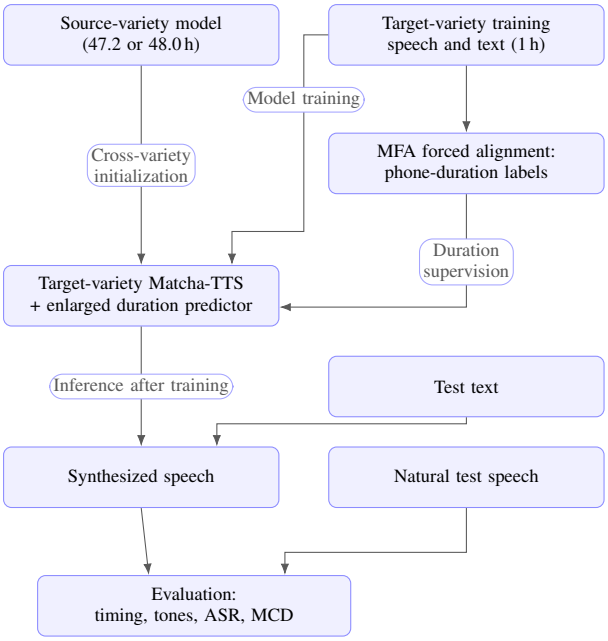


Fig. 1. Overview of XAD-TTS.

III. Experimental Setup

A. Data

The experiments use the TTS part of the Hakka Across Taiwan (HAT) corpus [10]: read speech in Sixian and Hailu, each by one professional female and one professional male speaker. Validation and test sets are defined by recording prompt, so their prompts never occur in training. Training subsets of 1 h and 10 h are balanced across the two speakers and nested (the 10 h set contains the 1 h set). Pinyin is split deterministically into initial, vowel nucleus, and coda, and the tone is attached to the nucleus symbol (“joint” symbols). Section IV-A also uses “separate” symbols, in which tones are

separate inputs, so 84% instead of 55% of the symbols are shared by the two varieties (symbols used by both varieties divided by symbols used by either).

B. Systems and training

Three systems are compared, all with joint symbols: Base, the original Matcha-TTS baseline trained from scratch with MAS durations; +AD, trained from scratch with alignment-supervised durations (AD) only; and XAD, the full XAD-TTS recipe, which adds cross-variety initialization and the enlarged predictor to +AD. Each is trained with Adam (learning rate 10^{-4}, batch 16) for a fixed 40,000 steps with three training seeds (1234, 123, 2026) at 1 h and 10 h, and the 40,000-step checkpoint is evaluated. As a high-resource reference, Base is also trained on the full pool (48.0 h of Sixian, 47.2 h of Hailu) with early stopping and a single seed. One 40,000-step run takes about 2 h on a single RTX 5070 Ti GPU (about 5.3 steps/s, 3.9 GB memory), and Base has 20.9 M parameters. Synthesis uses 10 ODE steps, temperature 0.667, and the universal HiFi-GAN vocoder.

C. Evaluation

For each variety, 100 test sentences (50 per speaker) are synthesized. 95% confidence intervals (CIs) are obtained by resampling sentences 5,000 times; with several seeds, every seed uses the same resampled sentences.

Spectral distortion

Mel-cepstral distortion (MCD) [11] is computed on 24 WORLD [12] mel-cepstral coefficients (without c_0) after dynamic time warping and averaged over five generation seeds per sentence.

Segment timing

Each synthesized waveform is force-aligned with the same MFA model, and the duration of every vowel nucleus is compared with the same nucleus in the natural recording. Timing accuracy is the share of nuclei within ± 30 ms (2.6 frames) of the natural duration, reported separately for checked and other syllables, together with the Pearson correlation ρ of checked-syllable nucleus durations. Tolerances of ± 20 and ± 50 ms give the same ranking of Base, +AD, and XAD.

Tone recognition

A gradient-boosting classifier is trained on natural training speech with acoustic features only: a five-point F0 contour of the nucleus in semitones, its mean and slope, log-duration, voicing ratio, and the F0 of neighboring syllables. Its accuracy on the synthesized syllables is reported, and its accuracy on natural test recordings serves as a reference.

Speech recognition check

Omnilingual ASR [13] (1B LLM decoder, version 2, language code `hak_Latn`) transcribes Hakka in a romanization with tone diacritics and stop codas *-p*, *-t*, *-k*. Each natural recording and its synthetic versions are transcribed, and the transcripts are aligned word by word. The measures are the character error rate (CER) between the ASR transcript of synthesized speech and that of the corresponding natural recording, computed on Latin letters and tone marks with spaces, digits, and non-Latin characters removed and pooled over sentences, the retention of checked syllables (natural-transcript words ending in *-p*/*-t*/*-k* whose aligned synthetic word still does), the letter-similarity gap between checked and other words, and the tone-mark agreement of aligned words. Twenty of the 200 test sentences (12 Sixian, 8 Hailu) are excluded because the recognizer's output for the natural recording was mostly non-Latin text; the remaining 180 sentences are the same for all systems, and the results are pooled over the two varieties. Copy-synthesis, in which natural mel spectrograms are passed through the same vocoder, serves as a reference condition.

IV. Results

A. Cross-variety initialization

In a preliminary sweep over five data sizes from 1 to 48 h, initialization lowered the validation loss in 16 of 20 settings; the gain was largest at 1 h (0.08–0.13) and was close to zero under the full-data conditions (48.0 and 47.2 h), so the analysis focuses on 1 h. Table I compares MCD between training from scratch and cross-variety initialization with 1 h of target-variety data and one training seed. Each model is trained for 40,000 steps with validation every 1,000 steps, and the checkpoint with the lowest validation loss is evaluated; this differs from Section IV-B, where the 40,000-step checkpoint is evaluated. Δ is the MCD of the initialized model minus that of the model trained from scratch, computed per sentence from unrounded values, so it can differ by 0.01 from the difference of the rounded means. The 95% CI in brackets is obtained by resampling sentences 10,000 times within each speaker.

TABLE I
Effect of Cross-Variety Initialization on MCD.

Target	Symbols	Scratch	Init.	Δ [95% CI]
Sixian	joint	6.36	5.99	-0.36 [-0.39, -0.34]
Sixian	separate	6.17	6.08	-0.09 [-0.12, -0.06]
Hailu	joint	6.46	6.59	+0.13 [+0.09, +0.18]
Hailu	separate	6.48	6.22	-0.26 [-0.30, -0.22]

Initialization lowers MCD in three of four settings, by up to 0.36 dB, but raises it for Hailu with joint symbols (+0.13 dB), mostly for the male speaker (+0.35 dB vs. -0.08 dB). Initialization alone is therefore helpful but not uniformly so, and XAD-TTS pairs it with alignment supervision.

B. Main results at one hour

Table II summarizes the results. Checked and Other give the timing accuracy of checked and other syllables, ρ the duration correlation of checked syllables, ΔMCD the paired difference from Base (1 h), and Tone the tone-classification accuracy. The 1 h system results are averaged over three training seeds. The full-data Base systems, Base (48.0 h) and Base (47.2 h), use one training seed, while the natural-recording rows report the classifier accuracy on real speech.

TABLE II
RESULTS FOR THE 1 H SYSTEMS AND FULL-DATA REFERENCES.

System	Checked (%) ↑	Other (%) ↑	ρ ↑	ΔMCD (dB) ↓	Tone (%) ↑
Sixian					
Base (1 h)	71.0	63.7	0.535	—	64.3
+AD (1 h)	77.0	70.4	0.651	−0.12	69.0
XAD (1 h)	**81.3**	**74.0**	**0.738**	**−0.33**	**73.0**
Base (48.0 h)	81.1	73.9	0.767	—	—
Natural recordings	—	—	—	—	71.5
Hailu					
Base (1 h)	69.4	54.9	0.576	—	62.1
+AD (1 h)	77.7	58.9	0.676	−0.10	64.2
XAD (1 h)	**82.2**	**65.9**	**0.743**	**−0.24**	**67.3**
Base (47.2 h)	83.0	68.6	0.797	—	—
Natural recordings	—	—	—	—	67.1

Both +AD and XAD improve the measured duration accuracy over Base, and XAD provides further gains over +AD under the evaluated settings. Alignment supervision alone (+AD) raises the share of checked-syllable vowels within ±30 ms by 6.0 (Sixian) and 8.3 (Hailu) percentage points. XAD adds another 4.3 and 4.5 points, reaching 81.3% [77.6, 85.2] and 82.2% [78.0, 86.4] against 71.0% [66.9, 75.2] and 69.4% [65.3, 73.8] for Base, which is close to the high-resource references (81.1% and 83.0%) for this measure. Other syllables improve as well (63.7% to 74.0% and 54.9% to 65.9%), and the correlation of checked-syllable durations rises from 0.54 to 0.74 (Sixian) and from 0.58 to 0.74 (Hailu), still below the references (0.77 and 0.80). XAD lowers MCD by 0.33 [0.31, 0.35] and 0.24 [0.20, 0.28] dB, two to three times the reduction from +AD (0.12 and 0.10 dB), and all three seeds move in the same direction. Tone-classification accuracy rises by 8.8 [7.7, 9.9] and 5.3 [4.4, 6.2] percentage points (paired differences pooled over seeds and syllables, so they can differ by 0.1 from the differences of the rounded means in Table II), and XAD exceeds +AD by 4.1 and 3.2 percentage points. The classifier accuracy on XAD-generated speech is close to that on natural recordings (measured on all test sentences). This does not imply that the synthesized tones are more natural or more accurate than human speech.

C. Effect of data size

Fig. 2 compares checked-syllable timing accuracy at 1 h and 10 h, with three seeds each, against the single high-resource Base model; the 95% CIs of the plotted values are about ±4 points.

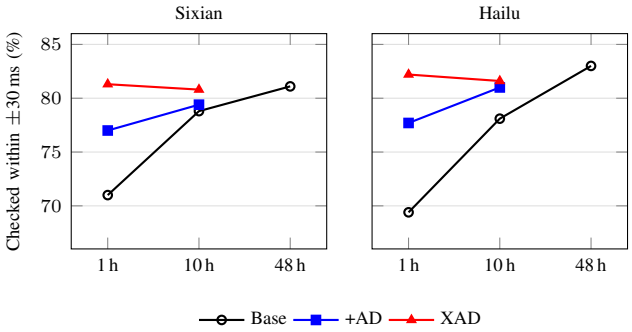


Fig. 2. Checked-syllable timing accuracy versus amount of training data.

The gains shrink with more data. At 10 h, XAD improves the checked-syllable share by 2.0 (Sixian) and 3.5 (Hailu) points and MCD by 0.24 and 0.16 dB, while +AD alone no longer lowers MCD (+0.04 and +0.03 dB). The recipe is therefore most useful in the regime it targets, when only about an hour of speech is available; there, XAD with one hour of data comes close to the checked-syllable timing of the full-data baselines.

D. Speech recognition check

Table III lists the speech recognition measures pooled over both varieties (180 sentences), including the copy-synthesis reference; Gap is the letter similarity of checked words minus that of other words.

TABLE III
ASR-BASED EVALUATION RESULTS.

System	CER (%) ↓	Checked kept (%) ↑	Gap ↑	Tone marks (%) ↑
Base (1 h)	22.2	77.8	−4.1	73.9
+AD (1 h)	21.8	81.5	−1.5	76.7
XAD (1 h)	21.7	**82.6**	**−0.5**	77.1
Base (47.2/48.0 h)	19.6	80.1	−3.7	77.2
Copy-synthesis	9.8	89.4	+0.2	88.9

The recognizer never saw the aligner or the tone classifier, so it gives an independent view. Overall CER does not separate the 1-h systems (−0.5 [−2.7, +1.9] points for XAD), while the high-resource baseline is better (−2.6 [−4.5, −0.5]). The differences appear where XAD targets them. Checked syllables keep their stop coda more often (+4.8 [1.6, 8.0] points), the handicap of checked words relative to other words shrinks from −4.1 to −0.5 points (+3.6 [1.2, 6.0]), and tone marks agree more often with those of natural speech (+3.2 [1.9, 4.4]). The high-resource baseline improves tone marks equally but leaves the checked-word handicap almost unchanged (−3.7), which suggests that the checked-syllable gain comes from duration supervision rather than from more data. With +AD alone the same pattern holds with smaller gains (+3.7, +2.6, and +2.8 points). Copy-synthesis reaches 89.4% retention, a gap of +0.2, and 88.9% tone agreement. XAD thus removes the checked-word handicap and closes about 40% of the retention gap between Base and this reference, while the large

CER difference (9.8% vs. 21.7%) shows that most of what remains is general acoustic quality rather than timing.

E. Negative results

Inference-time duration scaling, dropout in or pre-training of the duration predictor, replaying source-variety data during target training, and averaging the durations of several seeds gave no further gain over XAD.

V. Discussion and Limitations

At one hour, alignment supervision accounts for most of the timing gain (6.0 of 10.3 percentage points for Sixian, 8.3 of 12.8 for Hailu), while cross-variety initialization with the larger predictor contributes most of the spectral gain and about half of the tonal gain. The remaining vowel-duration errors concentrate before pauses, where the predictor lengthens vowels more than the speakers do. The study has limitations. A listening test has not yet been completed, so all results are objective; a paired-comparison listening test is under way. All systems share one architecture and vocoder. The recipe needs a pronunciation lexicon and an in-domain aligner; here the aligner was trained on all recordings of each variety, i.e., on more target-variety audio than the one-hour TTS training set. The high-resource reference is a single model. The speech recognizer and the tone classifier are imperfect judges; the classifier reaches only 71.5% and 67.1% on natural speech.

VI. Conclusion

XAD-TTS combines cross-variety initialization with alignment-supervised durations and an enlarged duration predictor. With one hour of Hakka speech, it brings checked-syllable timing close to that of a baseline trained on about 48 hours, reduces spectral distortion, and improves objective tone recognition in both Sixian and Hailu, and a multilingual recognizer shows better agreement with natural speech on checked-syllable codas and tone marks. The recipe adds no inference-time component and trains in about two hours on a single consumer GPU, which makes it a practical starting point for voices in other low-resource varieties.

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References

[1] L.-W. Chen, H.-S. Lee, and C.-C. Chang, "VoxHakka: A dialectally diverse multi-speaker text-to-speech system for Taiwanese Hakka," arXiv:2409.01548, 2024.

[2] S. Mehta, R. Tu, J. Beskow, É. Székely, and G. E. Henter, "Matcha-TTS: A fast TTS architecture with conditional flow matching," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2024.

[3] J. Kim, S. Kim, J. Kong, and S. Yoon, "Glow-TTS: A generative flow for text-to-speech via monotonic alignment search," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[4] Y.-J. Chen, T. Tu, C. chieh Yeh, and H. yi Lee, "End-to-end text-to-speech for low-resource languages by cross-lingual transfer learning," in *Proc. Interspeech*, 2019.

[5] J. Xu, X. Tan, Y. Ren, T. Qin, J. Li, S. Zhao, and T.-Y. Liu, "LRSpeech: Extremely low-resource speech synthesis and recognition," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining (KDD)*, 2020.

[6] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu, "FastSpeech 2: Fast and high-quality end-to-end text to speech," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.

[7] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, "Montreal Forced Aligner: Trainable text-speech alignment using Kaldi," in *Proc. Interspeech*, 2017.

[8] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.

[9] J. Kong, J. Kim, and J. Bae, "HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[10] Hakka Affairs Council, "Hakka across Taiwan (HAT) speech corpus," Distributed by the Association for Computational Linguistics and Chinese Language Processing (ACLCLP), catalog no. 2024-005, 2024.

[11] R. F. Kubichek, "Mel-cepstral distance measure for objective speech quality assessment," in *Proc. IEEE Pacific Rim Conf. Communications, Computers and Signal Processing*, 1993.

[12] M. Morise, F. Yokomori, and K. Ozawa, "WORLD: A vocoder-based high-quality speech synthesis system for real-time applications," *IEICE Trans. Inf. & Syst.*, vol. E99-D, no. 7, pp. 1877–1884, 2016.

[13] Omnilingual ASR team, "Omnilingual ASR: Open-source multilingual speech recognition for 1600+ languages," arXiv:2511.09690, 2025.